

BRAIN TUMOR IDENTIFICATION

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OVERVIEW

Manually analyzing and identifying a brain tumor is a slow and labor-intensive process that runs the possibility of misinterpretation due to a medical professional's fatigue or even lack of time. By developing and properly training an image classification neural network learning algorithm, it may be possible to identify and classify brain tumors leading to prompt diagnosis and treatment.

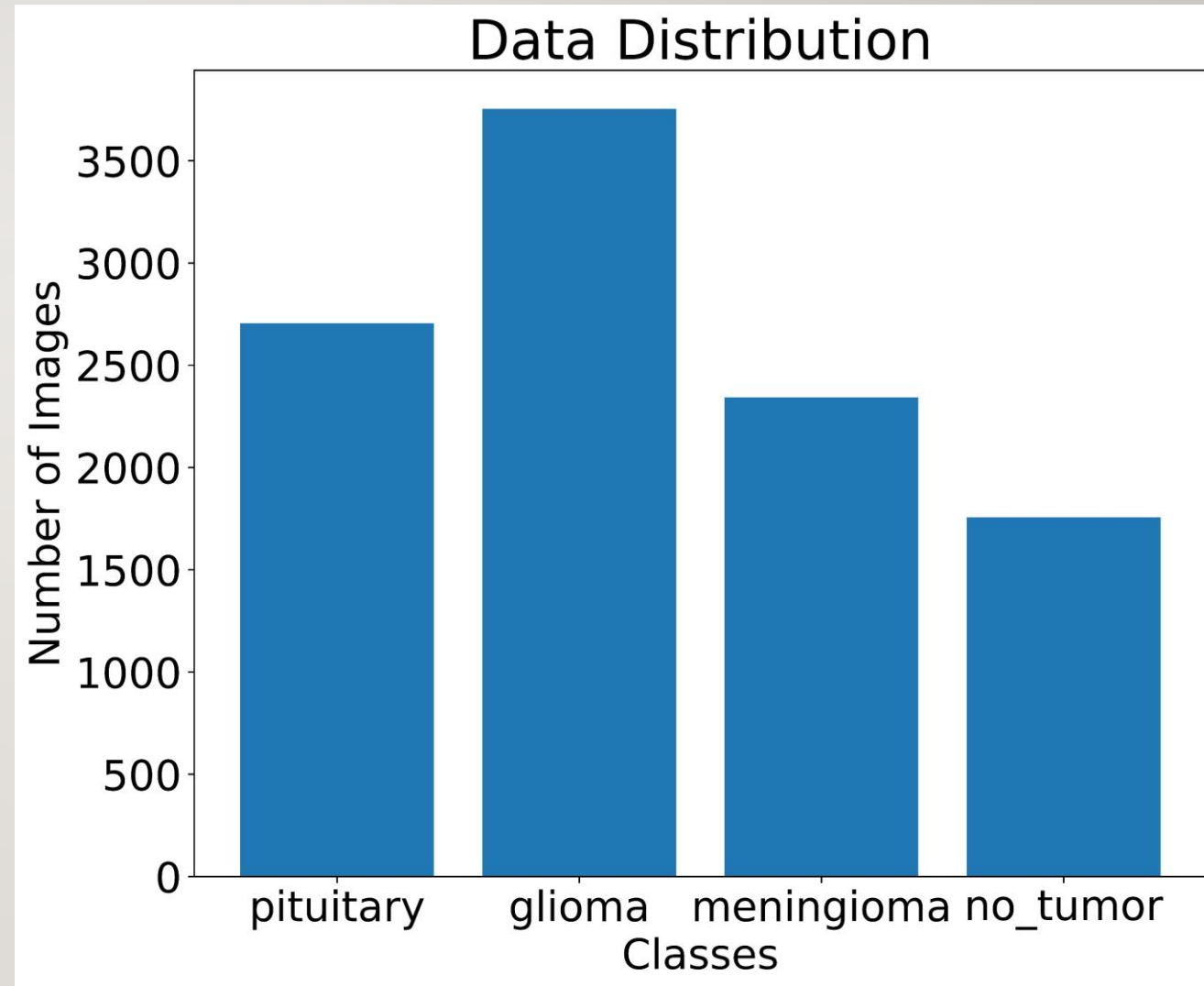
- Prompt diagnosis
- Brain tumor identification/classification
- Model Results: 98% recall average

BUSINESS UNDERSTANDING

- Problem: Slow diagnosis or misinterpretation
- Goal: Brain tumor identification/Classification

DATA UNDERSTANDING

- Data from Kaggle Datasets
- About the data
- Data Distribution



MODEL

- Neural Network
- DenseNet model

EVALUATION

- Model performance
- Confusion Matrix

True Labels	Predicted Labels			
	glioma	meningioma	no_tumor	pituitary
glioma	553	9	2	0
meningioma	9	341	1	1
no_tumor	0	0	265	0
pituitary	1	0	1	405

RECOMMENDATION

- Model implementation
- Gather more data

NEXT STEPS

- Model fine tuning
- Cat scan testing

THANK YOU

